

Preliminary Design Review

Objective

The purpose of this activity is to conduct an initial review of your design to make sure it meets the requirements specified in the 'Design Requirements Analysis' coursework. In the following descriptions, the word 'robot' means the mobile manipulator (Leo Rover + Manipulator) and all its peripherals.

The Preliminary Design Review coursework should have the following structure. All sections should have appropriate contextualisation text (i.e., don't just put the images in).

- **Introduction (-5% if missing)**
 - A summary of the problem statement and objectives.
- **System (10%)**
 - System block-diagram of all the hardware components in your robot.
 - Ensure all the components that have been used are described and justified as to why they are needed.
- **Mechanical Design (25%)**
 - 3D CAD model of your robot.
 - Design files for your payload sled.
 - Discussion of any design trade-offs or challenges with component placement.
 - Discussion about manufacturability and suitability of your sled design (e.g. strength, stability, etc...)
- **Electrical Design (10%)**
 - Power connection diagram for all the peripherals on your robot.
 - This could be included as part of the system block diagram, but it should include values and units
 - Power Budget
 - Calculate if you have enough power to run everything.
- **Software Design (15%)**
 - Software Block Diagram
 - Include where the software will reside (i.e. NUC or RPi)
 - A link to your Team's Git repository
 - Documentation of your code repository will be evaluated.
- **Analysis (30%)**
 - Using the Requirements Verification Matrix, you should analyse your design meets the requirements.
- **Project Plan (10%)**
 - A project plan (Gantt Chart) with clear work packages, resource allocations (who is doing what), milestones and deliverables for S2.
- **Peer Review**
 - All students must complete the Peer Review assessment on Blackboard.
- **Appendix (-10% if missing)**
 - Updated Design Requirements Document
 - Updated Workplace Charter

Submission

There will be an assignment submission link on Canvas. This assessment is a 'Group' assessment, however the mark awarded will be scaled for each individual student based on the Peer Review 2 weightings. **One student from each Team must complete the assignment by the deadline.**

The submission deadline is **Friday 12th December 14:00**.

Mitigating Circumstances

Any mitigating circumstance requests should be directed to the Teaching Office.

SAW

Issue 1.4

10/07/2025